

History of Microbiology

Definition, Scope, Origin of Microbial Life and Spontaneous Generation

Lecture Roadmap

- ① Definition of microbiology
- ② Scope and applications
- ③ Historical development of microbiology
- ④ Origin of microbial life
- ⑤ Theory of spontaneous generation
- ⑥ Experiments disproving spontaneous generation
- ⑦ Biogenesis, germ theory, sterilization and aseptic techniques

Learning Outcome

Students should connect historical discoveries with modern laboratory microbiology and biotechnology practice.

2. Definition of Microbiology

Definition

Microbiology is the branch of biology that studies microorganisms, including their structure, growth, metabolism, genetics, ecology, pathogenicity and applications.

Word origin

- **Mikros** = small
- **Bios** = life
- **Logos** = study

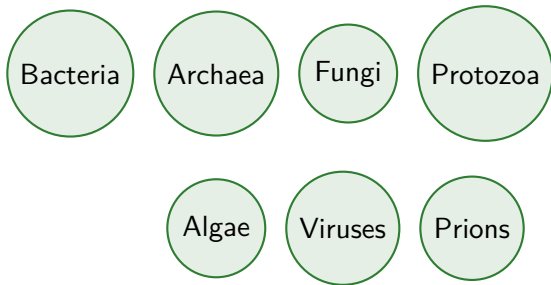
Simple meaning

- Study of microscopic life
- Study of organisms too small to be seen clearly by the naked eye

Example

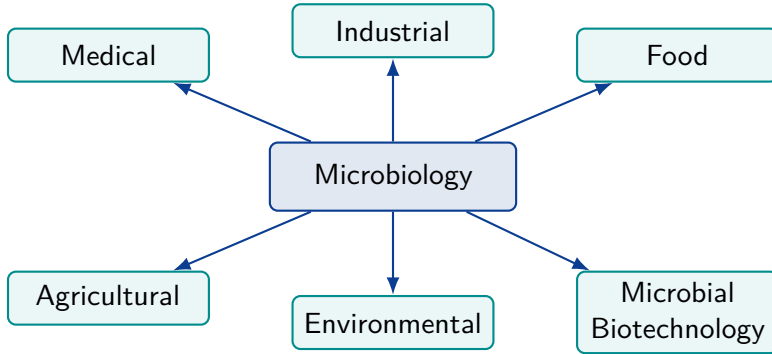
Curd formation, food spoilage, fermentation, infection and decomposition are all microbial activities.

Microorganisms Studied in Microbiology



Group	Example
Bacteria	<i>Escherichia coli</i> , <i>Bacillus subtilis</i>
Fungi	Yeast, moulds, <i>Penicillium</i>
Protozoa	<i>Amoeba</i> , <i>Paramecium</i>
Algae	<i>Chlamydomonas</i> , diatoms
Viruses	Influenza virus, bacteriophage

3. Scope of Microbiology



Key idea

Microorganisms are found in soil, water, air, food, human body, plants, animals and extreme environments.

3.1 Medical Microbiology

Meaning

Medical microbiology studies microorganisms that cause human diseases and methods to diagnose, prevent and treat infections.

Disease	Causative agent
Tuberculosis	<i>Mycobacterium tuberculosis</i>
Cholera	<i>Vibrio cholerae</i>
Typhoid	<i>Salmonella Typhi</i>
Malaria	<i>Plasmodium</i> species
COVID-19	SARS-CoV-2

Classroom example

When a patient has typhoid fever, laboratory culture, biochemical tests and serological tests help identify *Salmonella Typhi*.

3.2 Industrial Microbiology

Meaning

Industrial microbiology uses microorganisms for large-scale production of useful products.

Product	Microorganism
Alcohol	<i>Saccharomyces cerevisiae</i>
Citric acid	<i>Aspergillus niger</i>
Penicillin	<i>Penicillium</i>
Enzymes	<i>Bacillus, Aspergillus</i>

Biotechnology link

Microbes are used as biological factories for:

- antibiotics
- enzymes
- vaccines
- recombinant proteins

3.3 Food Microbiology

Useful roles

- Curd formation
- Cheese production
- Bread making
- Fermented foods
- Probiotics

Harmful roles

- Food spoilage
- Food poisoning
- Toxin production
- Contamination during storage

Example

Lactobacillus converts lactose into lactic acid during curd formation.

3.4 Agricultural Microbiology

Meaning

Agricultural microbiology studies microorganisms associated with soil fertility, plant growth and plant disease management.

Function	Microbial example
Nitrogen fixation	<i>Rhizobium</i> , <i>Azotobacter</i>
Phosphate solubilization	<i>Bacillus</i> , <i>Pseudomonas</i>
Biocontrol	<i>Trichoderma</i> , <i>Bacillus thuringiensis</i>
Composting	Bacteria and fungi

Example

Rhizobium forms nodules in legume roots and fixes atmospheric nitrogen.

3.5 Environmental Microbiology

Meaning

Environmental microbiology studies microbial roles in natural ecosystems and pollution control.

Major roles

- Decomposition
- Carbon cycle
- Nitrogen cycle
- Sulphur cycle
- Wastewater treatment
- Bioremediation

Example

Oil-degrading bacteria can break down hydrocarbons and help clean oil-contaminated environments.

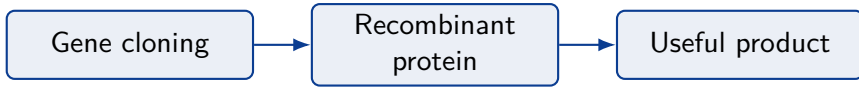
Why important?

Microbes recycle nutrients and maintain ecological balance.

3.6 Microbial Biotechnology

Meaning

Microbial biotechnology uses microorganisms or microbial products for useful applications in medicine, agriculture, industry and research.



Applications

- Insulin production
- Vaccine development
- Enzyme production
- Biofuel production

Example

Genetically modified *E. coli* can produce human insulin.

4. Importance of Studying Microbiology History

Why history matters

History shows how scientific knowledge develops through observation, experimentation, debate and evidence.

Before microorganisms were discovered, people could observe microbial effects but could not explain them.

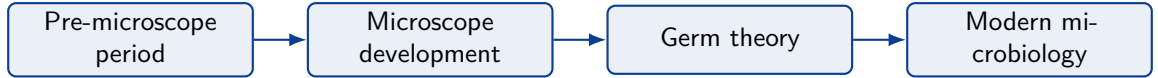
Observed effects

- Fermentation
- Food spoilage
- Disease spread
- Decomposition
- Wound infection

Earlier explanations

- Bad air
- Supernatural causes
- Imbalance of body fluids
- Spontaneous generation

5. Historical Development of Microbiology



Key transition

Microbiology changed from speculation to experimental science after microscopes, culture methods, staining and sterilization techniques were developed.

5.1 Pre-Microscope Period

Main idea

Before microscopes, microorganisms were unknown, but their activities were noticed in daily life.

Common observations

- Milk turning into curd
- Grapes fermenting into wine
- Meat decay
- Spread of epidemics
- Wound infection

Example

People saw maggots appear on decaying meat and believed they arose from the meat itself.

Result: This led to belief in spontaneous generation.

5.2 Development of the Microscope

Robert Hooke

- Published *Micrographia* in 1665
- Observed cork sections
- Described small compartments as “cells”

Antonie van Leeuwenhoek

- Made simple microscopes
- Observed pond water, rainwater and dental plaque
- Called microbes “animalcules”
- Known as Father of Microbiology

Teaching example

Dental plaque contains bacteria. Leeuwenhoek was among the first to observe such microscopic living organisms.

6. Major Contributors in Microbiology

Scientist	Major contribution
Robert Hooke	Described cells in cork
Antonie van Leeuwenhoek	Observed bacteria and protozoa as animalcules
Francesco Redi	Challenged spontaneous generation using meat and maggots
John Needham	Supported spontaneous generation using boiled broth experiment
Lazzaro Spallanzani	Opposed spontaneous generation using improved broth experiment
Louis Pasteur	Disproved spontaneous generation using swan-neck flask experiment
Joseph Lister	Introduced antiseptic surgery
Robert Koch	Developed Koch's postulates and pure culture methods
Edward Jenner	Developed smallpox vaccination
Alexander Fleming	Discovered penicillin

Exam point

Pasteur's experiment is one of the most important experiments in the foundation of modern microbiology.

7. Origin of Microbial Life

Basic idea

The first life forms on Earth were probably simple, unicellular, anaerobic prokaryotic microorganisms.

Early Earth conditions

- High temperature
- Volcanic activity
- Lightning
- UV radiation
- Little or no free oxygen

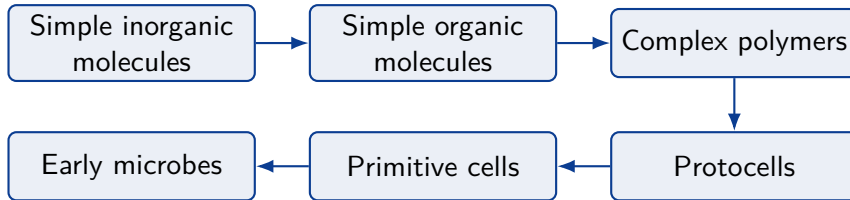
Probable first cells

- Anaerobic
- Prokaryotic
- Unicellular
- Chemotrophic or heterotrophic
- Adapted to harsh environments

7.2 Chemical Evolution

Chemical evolution

Simple inorganic molecules gradually formed organic molecules, which later gave rise to primitive cells.



Important distinction

Abiogenesis explains the origin of first life over long evolutionary time. It is not the same as spontaneous generation.

7.3 Oparin–Haldane Hypothesis

Hypothesis

Life originated from simple chemicals under early Earth conditions through gradual chemical evolution.

- 1 Early Earth had a reducing or low-oxygen atmosphere.
- 2 Energy from lightning, UV radiation and volcanic heat caused reactions.
- 3 Simple organic molecules formed.
- 4 Organic molecules accumulated in oceans as “primordial soup”.
- 5 Larger polymers and membrane-like structures formed.
- 6 Primitive cells gradually evolved.

Example

Amino acids, sugars and nitrogenous bases are basic molecules needed for life.

7.4 Miller–Urey Experiment

Experimental idea

Stanley Miller and Harold Urey simulated early Earth conditions using gases and electric sparks.

Components

- Methane
- Ammonia
- Hydrogen
- Water vapour
- Electric spark as lightning

Result

Amino acids and other organic compounds were formed.

Significance

Organic molecules necessary for life can form naturally under suitable conditions.

7.5 First Microorganisms

Probable nature of first microorganisms

The first organisms were most likely simple prokaryotes that lived without oxygen.

Why anaerobic?

- Early atmosphere had little free oxygen
- Oxygenic photosynthesis evolved later
- Aerobic respiration developed after oxygen increased

Possible metabolism

- Fermentation-like pathways
- Chemotrophy
- Use of inorganic compounds
- Survival in extreme conditions

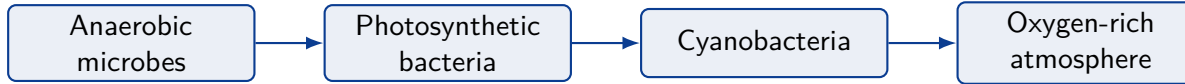
Example

Modern thermophiles and anaerobes help us understand possible early microbial lifestyles.

7.6 Photosynthetic Microorganisms

Key development

Some microorganisms evolved photosynthesis. Later, cyanobacteria evolved oxygenic photosynthesis.



Importance of cyanobacteria

Cyanobacteria released oxygen and contributed to the Great Oxygenation Event, supporting later evolution of aerobic life.

7.7 Endosymbiotic Theory

Theory

Mitochondria and chloroplasts originated from bacteria that lived inside primitive eukaryotic cells.

Mitochondria

- Originated from aerobic bacteria
- Helped cells use oxygen efficiently

Chloroplasts

- Originated from cyanobacteria
- Enabled photosynthesis in plants and algae

Evidence

Own DNA, 70S ribosomes, binary fission, double membrane and bacterial size.

8. Theory of Spontaneous Generation

Definition

Spontaneous generation was the old belief that living organisms could arise suddenly from non-living matter.

Observation	Old belief
Maggots on meat	Maggots arise from meat
Mice near stored grains	Mice arise from grains
Microbes in broth	Microbes arise from broth
Frogs in mud	Frogs arise from mud

Key point

This theory was accepted because contamination, eggs, larvae, spores and microbes were not understood.

8.2 Why People Believed It

- 1 Microbes were invisible to the naked eye.
- 2 Microscopes were not widely available.
- 3 Food spoiled without visible cause.
- 4 Flies and maggots appeared suddenly.
- 5 Microbial spores were unknown.
- 6 Sterilization was not understood.

Teaching example

If a piece of meat is left open, maggots appear after some time. Without knowing fly eggs, people assumed maggots came from meat.

Modern explanation

Maggots develop from eggs laid by flies.

9.1 Francesco Redi's Experiment

Aim

To test whether maggots arise spontaneously from meat.

Jar	Condition	Observation
Jar 1	Open jar	Maggots appeared
Jar 2	Sealed jar	No maggots
Jar 3	Covered with gauze	Maggots on gauze, not meat

Conclusion

Maggots came from fly eggs, not from meat. Redi challenged spontaneous generation for larger organisms.

9.2 John Needham's Experiment

Experiment

Needham boiled nutrient broth, poured it into flasks and sealed them. Later, microbial growth appeared.

Needham's conclusion

- Microorganisms arose spontaneously from broth.
- He believed a “vital force” produced life.

Limitation

The broth may not have been boiled long enough to kill all microbes and heat-resistant spores. Contamination may also have occurred.

9.3 Lazzaro Spallanzani's Experiment

Improved method

Spallanzani boiled broth for a longer time and sealed the flasks properly.

Observation

- No microbial growth in properly boiled and sealed flasks
- Growth occurred when flasks were opened

Conclusion

Microorganisms came from pre-existing microorganisms, not from the broth itself.

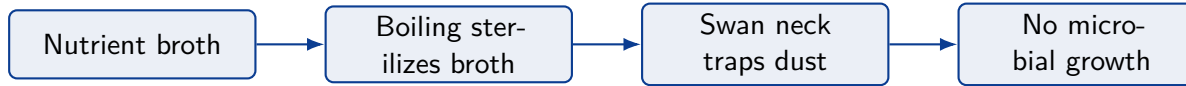
Criticism by supporters

They argued that boiling destroyed the “vital force” and sealed flasks prevented air from entering.

9.4 Louis Pasteur's Swan-Neck Flask Experiment

Purpose

To test whether microorganisms arise spontaneously or come from contamination in air and dust.



Key observation

Air could enter the flask, but dust and microbes were trapped in the curved neck. The broth remained sterile.

Pasteur's Experiment: Results and Conclusion

Condition	Result
Boiled broth in intact swan-neck flask	No microbial growth
Flask neck broken	Microbial growth occurred
Broth touched trapped dust in neck	Microbial growth occurred

Conclusion

Microorganisms came from pre-existing microorganisms in dust and air. They did not arise spontaneously from broth.

Principle established

Biogenesis: Life arises from pre-existing life.

10. Biogenesis vs Spontaneous Generation

Feature	Spontaneous generation	Biogenesis
Main idea	Life arises suddenly from non-living matter	Life arises from pre-existing life
Scientific status	Rejected	Accepted
Example	Maggots arise from meat	Maggots arise from fly eggs
Supported by	Needham's flawed experiment	Redi, Spallanzani and Pasteur
Importance	Historical idea	Basis of modern microbiology

Exam tip

Use Redi and Pasteur as strong experimental examples while explaining biogenesis.

11. Abiogenesis vs Spontaneous Generation

Feature	Abiogenesis	Spontaneous generation
Meaning	Origin of first life from non-living chemicals	Sudden formation of life from non-living matter
Time scale	Millions of years	Short period
Context	Early Earth conditions	Claimed to occur repeatedly to-day
Scientific value	Studied in origin-of-life science	Rejected historical theory
Example	Chemical evolution to first cells	Maggots from meat, microbes from broth

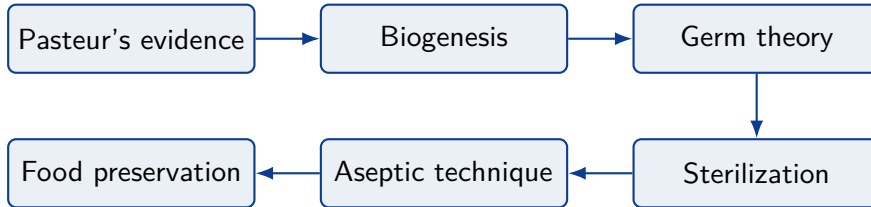
Simple memory line

Abiogenesis explains the first origin of life; spontaneous generation claimed repeated sudden origin of life.

12. Impact of Disproving Spontaneous Generation

Major scientific impact

Rejecting spontaneous generation changed biology, medicine, food preservation and biotechnology.



12.1 Development of Germ Theory

Germ theory

Germ theory states that specific microorganisms can cause specific diseases.

Connection with biogenesis

- Microbes do not appear spontaneously.
- They come from pre-existing microbes.
- Therefore, disease transmission can be explained.

Example

Tuberculosis is caused by *Mycobacterium tuberculosis*, not by bad air or spontaneous formation.

12.2 Development of Sterilization

Sterilization

Sterilization is the complete destruction or removal of all forms of microbial life, including spores.

Method	Example use
Boiling	Reduces microbial load in water
Autoclaving	Sterilizes media and glassware
Filtration	Sterilizes heat-sensitive solutions
Flaming	Sterilizes inoculating loop
Pasteurization	Reduces pathogens in milk

Laboratory example

Nutrient agar is usually autoclaved before pouring plates.

12.3 Development of Food Preservation

Why preservation became important

Understanding microbial contamination helped develop methods to prevent spoilage and foodborne disease.

Methods

- Pasteurization
- Canning
- Refrigeration
- Drying
- Salting
- Fermentation

Examples

- Pasteurization reduces pathogens in milk.
- Drying reduces water availability for microbes.
- Fermentation produces acids that inhibit spoilage organisms.

12.4 Development of Aseptic Techniques

Aseptic technique

Aseptic techniques are procedures used to prevent contamination by unwanted microorganisms.

Common practices

- Sterilize glassware and media
- Flame inoculating loop
- Work near flame or in laminar airflow
- Avoid unnecessary plate exposure
- Use sterile pipettes and tips

Classroom lab example

During bacterial streaking, the inoculating loop is flamed before and after touching the culture to avoid contamination.

Quick Revision

- 1 Microbiology studies microorganisms and their activities.
- 2 Microbes are useful in medicine, agriculture, food, environment and biotechnology.
- 3 Leeuwenhoek observed “animalcules” and is called Father of Microbiology.
- 4 Early life was probably anaerobic and prokaryotic.
- 5 Spontaneous generation was rejected by scientific experiments.
- 6 Pasteur's swan-neck flask experiment supported biogenesis.
- 7 Biogenesis led to germ theory, sterilization, food preservation and aseptic technique.

Student Activity / Discussion

Think and answer

- ① Why did Needham observe microbial growth in boiled broth?
- ② How did Pasteur allow air to enter but prevent microbial contamination?
- ③ Why is aseptic technique important in biotechnology laboratories?

Application question

A nutrient agar plate is left open for 30 minutes and colonies appear after incubation. Explain this using Pasteur's conclusion.

Thank You

Questions and Discussion

Microbiology begins with invisible life, but its impact is visible everywhere.